

DATA SCIENTIST
&
LEAD DATA ENGINEER

PARTHASARATHI SWAIN

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CAREER OBJECTIVE

- ✦ **Data Scientist | Data Engineer** with **11.4 years of proven experience**, delivering high-impact **AI-driven solutions** across **BFSI, E-commerce**, and **Energy & Utilities** domains. Adept at solving complex business challenges with a results-driven and innovative approach.
- ✦ Seeking a challenging role in a dynamic organization to leverage my expertise in state-of-the-art technologies, contribute to organizational growth, and continuously enhance my skills.

PROFESSIONAL SUMMARY

- ✦ **Oracle Certified Java Programmer (OCJP)** (v1.6) with an impressive score of 88%.
- ✦ **5+ years as a Data Scientist**, specializing in **Machine Learning, Deep Learning**, and **Generative AI** (XGBoost, Neural Networks, NLP, LLMs, OpenAI, Hugging Face). Expertise in **Regression, Classification**, and **Time Series Forecasting**.
- ✦ **7+ years as a Data Engineer**, with hands-on expertise in **Hadoop Architecture** (HDFS, Hive, Resource Manager) and building scalable **data pipelines** using **StreamSets, Control-M**, and **Databricks**.
- ✦ Skilled in optimizing AI/ML models and deploying **large-scale distributed systems** to drive operational efficiency and data-driven decision-making.

TECHNICAL SUMMARY

Skills	Experience
Data & AI Strategy	Machine Learning (Supervised/Unsupervised), Deep Learning, Advanced NLP, Computer Vision, LLMs, Generative AI
Statistics	Regression & Classification (Linear Regression, Logistic Regression, Decision Trees, Random Forest, SVM, XGBoost), Time Series Forecasting, RAG, Word2Vec, PCA, Ridge, Lasso
Visualization Tools	Tableau, Power BI
Programming Languages	Python, Java, Scala
Python Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch, NLTK, BeautifulSoup
Databases/NoSQL DB/DW	SQL, Oracle, MySQL, Cassandra, Snowflake
Big Data Technologies	Hadoop, HDFS, Hive, Yarn, Apache Kafka, Spark, PySpark, SparkSQL, Cloudera
Cloud Platform	AWS, Azure, Heroku
Deployment Tools	Git/GitHub, Streamlit, Flask, MLFlow, Docker, AWS Sagemaker, CI/CD Pipelines, DVC, BitBucket, MLOps
Project Methodology	Agile
Operating Systems	Windows, UNIX/Linux, CentOS, Ubuntu, RedHat

EDUCATION SUMMARY

- ✦ **B.Tech in Electronics & Telecommunication**, Biju Patnaik University of Technology, Odisha (2007-2011) with a CGPA of **8.01**.

INDUSTRY EXPERIENCE SUMMARY

VESTAS | Maintenance Cost Minimization & Active Power Forecasting For Wind Turbine
Data Scientist

<https://github.com/partha2018/maintenance-cost-minimization-for-wind-turbine>

July 2023 - Apr 2024
Bangalore

- ✦ Designed and implemented advanced solutions using **Large Language Models (LLMs)** with Hugging Face, OpenAI, and LangChain, enhancing model performance by 30%.
- ✦ Led **Generative AI projects**, aligning technical solutions with business goals to improve project efficiency by 25%.
- ✦ Built NLP projects using **LSTM and GRU**, improving linguistic model accuracy by 20% with effective prompt engineering and interface design.
- ✦ Developed a predictive application for **wind power output** using RNN models, achieving 95% accuracy in energy forecasting based on environmental data.
- ✦ Implemented **predictive maintenance models**, reducing maintenance costs by 30% and improving turbine uptime by 20%, saving \$4 million annually.

- ✚ Conducted exploratory data analysis on **sensor data**, identifying failure patterns that reduced maintenance costs by 15% through proactive interventions.
- ✚ Deployed machine learning models on **AWS SageMaker** and Databricks, reducing processing time by 40% and streamlining workflows with CI/CD pipelines.
- ✚ Created an AI-powered application integrating Hugging Face and OpenAI models to convert **images to audio**, enhancing data accessibility for maintenance insights.

GENPACT | Insurance Management System
Data Scientist

July 2021 – June 2023
Remote

<https://github.com/partha2018/insurance-management-system>

- ✚ Developed a **predictive model for insurance cost optimization** based on health and lifestyle data, improving pricing accuracy by 20% and reducing potential losses by 15%.
- ✚ Created **personalized insurance plans** tailored to individual health profiles, resulting in a 25% increase in customer satisfaction and retention.
- ✚ Built a **fraud detection system**, reducing claims costs by 15% and saving \$5 million annually through improved risk assessment.
- ✚ Designed and implemented **classification models** to identify potential customers, leveraging machine learning techniques to enhance customer acquisition.
- ✚ Streamlined data processing by automating **ETL pipelines**, achieving a 30% improvement in operational efficiency and input data quality for predictive models.
- ✚ Achieved a **95% predictive accuracy** for insurance cost models by applying advanced machine learning algorithms, hyperparameters tuning, and performance evaluation using metrics like MAE and R-squared.

DELOITTE | Inventory Management System
Data Scientist

May 2019 - May 2021
Bangalore

<https://github.com/partha2018/inventory-management-system>

- ✚ Developed a **Deep Learning-based inventory management system**, reducing storage costs by 15% and increasing sales opportunities by preventing stock-outs
- ✚ Designed a **transfer learning model** for efficient adaptation to company-specific data, reducing excess stock by 20% and optimizing inventory.
- ✚ Streamlined supply chain processes with optimization models, achieving a 25% reduction in inventory management costs and improving operational efficiency.
- ✚ Built and deployed **demand forecasting models** using historical sales and inventory data, reducing stockouts by 25% and enhancing predictive accuracy.
- ✚ Automated data pipelines with Streamsets and Python, cutting manual processing by 20% and improving data handling efficiency.
- ✚ Delivered an **Estimated Delivery Time prediction model** with 92% accuracy, boosting customer satisfaction through reliable delivery forecasts.

INTUIT | Uniform Customs and Practice for Documentary Credits (UCP)
Data Engineer

Nov 2018 – May 2019
Bangalore

- ✚ Optimized batch size in Streamsets, reducing job performance time from 45 minutes to 7 minutes, significantly enhancing operational efficiency.
- ✚ Designed and maintained robust data pipelines, performing data cleansing, aggregation, and transformation to ensure high-quality data flow.
- ✚ Developed data ingestion pipelines with Streamsets, processing 10 GB of big data daily and scheduling jobs using Control-M for timely execution.
- ✚ Leveraged Cassandra for staging, Tableau for visualization, Hive for querying, and Hadoop for storage, enhancing data processing capabilities with Cloudera.
- ✚ Executed complex SQL queries for big data analysis, enabling Auto-Scaling and End-time features in Databricks to optimize resource utilization.

ITC INFOTECH | Fusion Banking Essence (FBE)
Data Engineer

Apr 2017 – July 2018
Bangalore

- ✚ Developed scalable ETL pipelines using Streamsets, enhancing data processing efficiency by optimizing batch jobs and resolved Tungsten issues for over 10 applications.
- ✚ Designed and implemented data processing pipelines with Python and PySpark, streamlining customer eligibility data processing.
- ✚ Migrated data from an on-premises Cloudera environment to a cloud-based Snowflake data warehouse, increasing data accessibility and performance.

HCL TECHNOLOGIES | Converse (WMA)
Senior Software Engineer

Sep 2014 – Dec 2016
Bangalore

🚦 Worked as a **Senior Java Developer** for a leading **BFSI client**, delivering robust and scalable solutions.

ARICENT GROUP | Converse (WMA)
Software Engineer

Sep 2013 – Aug 2014
Bangalore

🚦 Worked as a **Java Developer** for a **BFSI client**, focusing on delivering reliable & efficient software solutions.

ACHIEVEMENTS

- 🚦 Over **8 years of teaching experience** in Data Science, serving as an AI & ML instructor at one of India's largest Ed-Tech platforms.
- 🚦 Recognized with the **Best Employee Award** and **Sports Award** during tenure at HCL Technologies.